(i) Freundlich adsorption isotherm.

According to Freundlich,

$$\frac{x}{m} = kp^{\frac{1}{n}}$$

where x is the amount of adsorbate, adsorbed on 'm' gm of adsorbent at a pressure of p. K and n are constant introduced by freundlich.

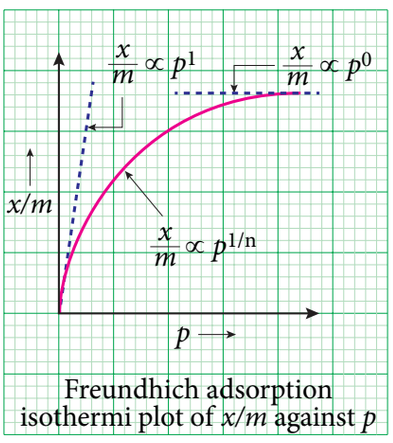
Value n is always less than unity.

This equation is applicable for adsorption of gases on solid surfaces. The same equation becomes $\frac{x}{m} = K\, c^{\frac{1}{n}}$, when used for adsorption in solutions with c as concentration.

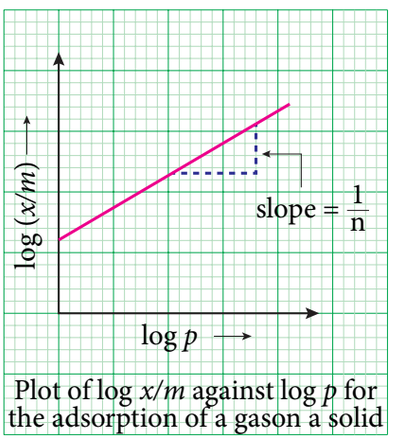
This equation quantitively predicts the effect of pressure(or concentration) on the adsorption of gases(or adsorbates) at constant temperature.

Taking log on both sides of equation $\frac{x}{m} = Kp^{\frac{1}{n}}$

$$\log \frac{x}{m} = \log K + \frac{1}{n} \log p$$



Freundhich adsorption isothermi plot of x/m against p



Plot of log x/m against log p for the adsorption of a gason a solid

Figure 10.2 $\log \frac{x}{m}$ vs log p **graph**

Hence the intercept represents the value of log k and the slope gives $\frac{1}{n}$.

This equation explains the increase of $\frac{x}{m}$ with increase in pressure. But experimental values show the deviation at low pressure.

Limitations

This equation is purely empirical and valid over a limited pressure range.

The values of constants k and n also found vary with temperature. No theoretical explanations were given.

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